

Power Analysis and Sample Size Planning

Advanced Topics: Methodology, Philosophy, and Research Design

1. What is Power?

Power (test strength) = probability of detecting an effect of a certain size as significant ($1 - \beta$).

Type II error (β): H_0 is retained when H_1 is true.

Convention: Power > 0.80 . Ethics committees and reviewers require power planning.

2. Factors Influencing Power

(1) Effect size (d, r, f): Larger effects = higher power.

(2) Sample size (n): Larger n = higher power.

(3) Alpha: $\alpha = .05 > .01$ = higher power.

(4) Test direction: One-tailed $>$ two-tailed.

(5) Design: Paired $>$ independent groups.

3. A Priori Power Analysis (Planning)

Given: Desired power ($.80$), α ($.05$), expected effect size. Find: n .

Effect size from: pilot studies, meta-analyses, or minimal relevant effect.

Conservative estimate: Better slightly larger sample than too small.

Tools: G*Power, `pwr` in R, `statsmodels` in Python.

4. Post-hoc Power (Retrospective)

Problematic: Post-hoc power is a 1:1 transformation of the p -value.

Alternative: Report confidence intervals instead.

5. Power for Different Methods

t-test (two independent): $n \sim 64$ per group for $d=0.5$, $\text{power}=.80$.

ANOVA, regression, chi-square: Use G*Power or dedicated functions.